

PRISM Manufacturing Intelligence Platform

Comprehensive Feature Audit & Value Assessment

Version: 9.0 (MCP Server v2.10.0) | **Date:** March 20, 2026 | **Classification:** CONFIDENTIAL

THE 4 FLAGSHIP PRODUCTS (Launch Priority)

PRODUCT 1: Ultimate Speed & Feed Calculator (SFC)

The most comprehensive speed/feed calculator in existence.

Accepts ANY subset of inputs and infers all missing parameters using physics, materials databases, and empirical lookup tables. Far exceeds Harvey Tool, Kennametal, and Sandvik calculators combined.

Core Engine: UltimateSpeedFeedEngine + 9 Sub-Engines

What makes it ultimate — accepts ANY input combination:

- Material name alone (e.g. "4140") resolves ISO group, hardness, Kienzle coefficients, Taylor constants
- Tool diameter alone infers flutes, coating, geometry, recommended chip load
- Just specify operation type and material — engine fills in everything else
- Every output parameter includes a confidence score (0-1) and the formula used

Physics Models (not empirical lookup tables like competitors):

- **Kienzle (1952):** $F_c = k_c 1.1 \times a_p \times f_z^{(1-m_c)}$ — real cutting force prediction
- **Taylor (1907):** $VT^n = C$ — tool life prediction with cost optimization
- **Loewen-Shaw:** Thermal damage risk with coating-aware temperature limits
- **Brammertz:** Surface finish prediction from feed geometry + runout + vibration
- **Gilbert:** Economic optimization balancing tool cost vs machine time
- **Chip Thinning:** Radial engagement compensation with empirical validation

7 Operations x 6 ISO Groups x 3 Cut Types = 126+ Parameter Combinations:

Operation	Capabilities
Milling	Face, profile, pocket, slot, adaptive, trochoidal, HSM, HPC, plunge
Turning	OD, ID, facing, grooving, threading, parting
Drilling	Standard, peck, gun-drill, deep hole
Tapping	Rigid, floating, thread milling
Reaming	Through, blind with finish optimization
Boring	Rough, finish with deflection compensation
Thread Milling	Single-point, multi-point with pitch compensation

3 Output Modes per Calculation:

- **Conservative:** Maximum tool life, minimum risk (new operators, expensive parts)
- **Balanced:** Optimal cost-per-part (default recommendation)
- **Aggressive:** Maximum productivity, MRR-optimized (high-volume production)

Machine Constraint Verification:

- Power budget check against machine_power_kw
- Torque verification against spindle curve
- RPM capping to machine_max_rpm
- Rigidity-adjusted depths for low/medium/high stiffness

Advanced Features (no competitor offers these):

- Stability lobe integration — avoids chatter RPM zones when stiffness/frequency provided
- Cost-per-part optimization with tool cost, change time, regrind data
- Runout impact analysis — predicts finish degradation from spindle/holder/tool TIR
- Edge radius ploughing force correction for micro-tools
- Coolant strategy adjustment factors (flood vs MQL vs cryogenic vs dry)

Backed by:

- 6,346+ materials with 127 Kienzle-enriched parameters each
- 95,608 tools from 28+ manufacturers with geometry data
- 888 machine profiles with spindle curves and power limits
- 499 manufacturing formulas across 27 domains
- 8+ MIT/Stanford university course algorithms

Fusion 360 Plugin Integration:

- FusionToolExportEngine: exports PRISM's 95K+ tools as Fusion 360 .tools JSON libraries
- FusionToolSyncEngine: smart partitioning (max 500 tools/library), sync state tracking, job-based export
- Auto-filled cutting parameters per material preset — tools arrive in Fusion with speeds/feeds pre-populated
- **Result: When users use our Fusion plugin, they get MORE ACCURATE speed/feed results** because we have real tool geometry, material-specific Kienzle coefficients, and machine-aware constraints

Plugin Expansion Roadmap:

- Mastercam (next) — tool library sync + parameter presets
- SolidCAM — iMachining parameter optimization
- HyperMill — strategy-aware speed/feed with material bridge
- NX/PowerMill/Esprit/GibbsCAM — planned

Estimated Value: \$120K-\$200K/yr per enterprise Savings: Replaces 3-5 standalone calculators (\$5K-\$15K/yr).

Reduces test cuts 50-70% (saving \$20K-\$80K/yr in scrap + machine time). Physics-based vs empirical means 15-25% better tool life from day one.

PRODUCT 2: Post Processor Generator (PPG)

Cross-CAM advanced post processing with custom algorithms, physics-backed feed optimization, and machine-specific feature injection.

Takes toolpath output from ANY CAM system and generates optimized G-code for ANY controller — with features no single CAM system offers alone.

Architecture: 3-Engine Pipeline

Engine 1 — PostProcessorEngine (Base Generation):

- 13 controller families: FANUC, HAAS, SIEMENS, HEIDENHAIN, MAZAK, OKUMA, BROTHER, HURCO, FAGOR, DMG_MORI, DOOSAN, MITSUBISHI, GRBL
- 20+ dialect variations within each family
- Canned cycle translation (G81/G83/G84/G73/G76 drilling + tapping + boring)
- Work offset management, safe start blocks, program structure

Engine 2 — AdvancedPostProcessorEngine (Feature Injection):

Feature	What It Does	Controllers
Adaptive Clearing	Injects constant-chip-load toolpath modifications with trochoidal fallback	All
HSM Arc Fitting	NURBS interpolation, corner rounding tolerance, jerk limiting	FANUC G05.1, SIEMENS COMPCAD, HAAS G187
Feed Optimization	Chip thinning compensation, corner slowdown, arc feed limits — PHYSICS-BACKED using Kienzle	All
Tool Management	Sister tooling rotation, break detection (probe/load/laser), auto wear offset	All
In-Process Measurement	Probe critical features every N parts, auto-compensate, SPC logging, alarm on OOT	All with probing
Multi-Axis RTCP	G43.4/G43.5 (FANUC), TRAORI (SIEMENS), TCPM (HEIDENHAIN), M128 (MAZAK)	5-axis controllers
Singularity Avoidance	Detects and linearizes near-pole moves in 5-axis	All 5-axis

Engine 3 — MasterPostProcessorEngine (Cross-CAM Orchestrator):

- Mix toolpaths from DIFFERENT CAM systems in one program
- Apply best features from each: iMachining constant load + HyperMill collision avoidance + Fusion adaptive
- Machine-specific feature codes from POST machine database (Haas G187, Okuma G08 smoothing, etc.)
- Tribal knowledge injection: 204+ tips from 5 CAM systems applied as post-processing enhancements
- RL-based format selection learns optimal output structure over time

G-Code Transpilation (17+ Dialects):

- Convert existing G-code between ANY controller dialects
- FANUC to SIEMENS, HAAS to HEIDENHAIN, etc.
- Canned cycle translation, coordinate system mapping, syntax transformation

Automatic Speed/Feed in Post (THE DIFFERENTIATOR):

- PostProcessorFeedOptimizerEngine recalculates feeds for corner dynamics
- AutoSpeedFeedEngine computes optimal S/F per operation segment
- Uses same UltimateSpeedFeedEngine physics for consistency
- **When combined with our Fusion plugin:** real tool geometry + real material data = feeds optimized beyond what any CAM system can do alone

Custom Algorithm Support:

- 6 novel PRISM algorithms: TGAR, HRAF, MTHZD, CFSF, PTDC, VCER
- 18 extended scientific algorithms
- Users can define custom feed profiles, smoothing rules, and cycle modifications
- Cross-CAM novel algorithms bring best practices from each platform

Estimated Value: \$200K-\$350K/yr per enterprise Savings: Replaces per-seat CAM post licenses (\$5K-\$15K each). Eliminates controller-specific post customization consulting (\$10K-\$50K per post). Feed optimization alone saves 10-20% cycle time across all programs.

PRODUCT 3: Print-to-Program Pipeline (Internal SFC + PPG)

Upload an engineering drawing → Get a complete CNC program with tool list, setup sheet, and confidence scoring. Uses Products 1 & 2 internally.

Five-stage pipeline from drawing intake to validated G-code output.

Pipeline: 5 Stages, 5 Specialized Engines

Stage 1 — Drawing Intake (BlueprintOCREngine + PDFBlueprintDimensionExtractorEngine)

- OCR extraction: dimensions (linear, diameter, radius, angular, thread, chamfer, counterbore, countersink)
- GD&T frame parsing: all 14 ASME Y14.5 symbols with datum references
- Title block: part number, revision, material, finish, scale, units
- Process notes classification: material, finish, safety, inspection categories
- Confidence scoring on every extracted element
- DXF text-layer dimension extraction for CAD files
- Revision comparison and change detection

Stage 2 — Feature Extraction & Classification (PrintToGeometryEngine)

- Converts extracted dimensions into machinable features
- 18 feature types: holes (through/blind/counterbore/countersink), pockets (open/closed), slots, contours, faces, steps, chamfers, fillets, threads (internal/external), bores, grooves, keyways
- 40+ specialized feature types including aerospace (turbine blade slots), medical (bone screw threads), mold (cooling channels, ejector pins)
- Generates parametric CadQuery Python scripts for 3D model creation
- GD&T requirements mapped to inspection methods

Stage 3 — Process Planning (Internal SFC + Tool Selection)

- UltimateSpeedFeedEngine calculates speeds/feeds for EVERY operation
- Automatic tool selection from 95,608-tool catalog
- Operation sequencing: rough → semi-finish → finish with proper ordering
- Machine constraint verification against selected machine profile
- Kienzle force, Taylor tool life, deflection checks on every operation
- Coolant strategy selection per operation

Stage 4 — Program Generation (Internal PPG)

- Full G-code output with safe start blocks, tool changes, coolant commands
- Controller-specific syntax via PostProcessorEngine (13 families)
- Advanced features injected via AdvancedPostProcessorEngine
- Feed optimization via PostProcessorFeedOptimizerEngine
- Canned cycles for drilling, tapping, boring operations

Stage 5 — Validation & Output

- Safety validation: collision checks, spindle limits, workholding verification
- Setup sheet generation with critical dimensions highlighted
- Tool list with holder/assembly recommendations
- Inspection plan from extracted GD&T
- Overall confidence score (0-1) for the complete program

- Warnings for any assumptions made

Blueprint-to-Quote Bridge (Xometry Competitor):

- BlueprintToQuoteBridgeEngine translates extracted data directly to QuoteEstimator input
- Auto-maps 20+ material callouts to quote parameters
- Detects secondary ops from drawing notes (heat treat, plating, NDT, etc.)
- Generates instant RFQ response — competes directly with Xometry's upload-to-quote

Estimated Value: \$150K-\$250K/yr per enterprise Savings: Eliminates 2-4 hours manual programming per new job.

For a shop doing 50 new jobs/month, that's 100-200 hours/month saved (\$5K-\$15K/month). First-part-right rate improvement from physics-backed parameters reduces scrap 30-50%.

PRODUCT 4: Full ERP / Business Management / Accounting

Complete shop management system — quoting, job tracking, scheduling, inventory, HR/payroll, customers, purchasing, financial reporting, and compliance. Replaces JobBOSS, E2, ProShop, and QuickBooks combined.

193 Actions Across 20+ Business Engines

Quoting & Cost Estimation (6 specialized engines):

Engine	What It Does
QuoteEstimatorEngine	Physics-backed quote generation with material, labor, machine time, markup
CostEstimationEngine	Manufacturing cost breakdown: setup, run, material, tooling, overhead
ActualCostEngine	Roll-up of actual costs vs quoted for margin tracking
SheetMetalQuoteEngine	Sheet metal specific: nesting, bend deductions, forming time
InjectionMoldQuoteEngine	Mold cost amortization, cycle time, material fill
AdditiveQuoteEngine	AM build time, material volume, support removal, post-processing
CastingQuoteEngine	Pattern cost, pour weight, finishing allowance
WeldFabQuoteEngine	Weld length, prep time, fixture, NDT costs
MultiProcessQuoteEngine	Unified quoting across any combination of processes
QuoteAnalyticsEngine	Win/loss tracking, quote-to-actual variance, margin analysis

Job Lifecycle Management:

- Job creation with full BOM, routing, and operations
- State machine: Quoted → Ordered → Scheduled → In-Progress → QC → Shipped → Invoiced → Closed
- Real-time status dashboards with KPI tracking
- Work order generation with operation sequences

Production Scheduling (5 algorithms):

- FIFO, SPT, EDD, CR, SLACK dispatching rules
- Johnson's algorithm for 2-machine flow shop
- Critical Path Method (CPM) for complex jobs
- Bottleneck identification and theory of constraints
- Capacity planning with machine load balancing
- Lead time estimation and due date tracking

Inventory Management:

- EOQ (Economic Order Quantity) optimization
- Safety stock calculations with demand variability
- ABC analysis for classification

- Tool inventory with reorder alerts
- Material tracking with lot/heat traceability

HR & Payroll (16+ actions):

- Employee lifecycle: create, search, skill matrix tracking
- Time clock: clock in/out, timecard management, overtime rules
- Payroll calculation from timekeeping with tax brackets
- Benefits, training records, compliance tracking
- Skill-based job assignment recommendations

Customer Management (14 actions):

- Customer CRUD with contact management
- Credit checking and terms management
- Communication logging (calls, emails, meetings)
- Sales pipeline: opportunity tracking through close
- Customer analytics: top customers, revenue trends, churn risk

Financial Reporting & Accounting:

- General Ledger with double-entry accounting
- Financial analysis: NPV, IRR, breakeven, machine ROI
- Job profitability reports with actual vs quoted variance
- Revenue/expense dashboards
- Stripe billing integration for SaaS deployment
- Invoice generation and accounts receivable

Purchasing & Supply Chain:

- Vendor directory with manufacturer catalogs (MSC, Grainger, etc.)
- Purchase order generation and tracking
- Supplier analytics and performance scoring
- Material price tracking and cost optimization

Reporting & Export:

- PDF/Excel/CSV/DXF report generation
- Production summaries, quality reports, financial dashboards
- Pareto analysis, waterfall charts, control charts
- Setup sheets, tool lists, inspection plans
- KPI dashboards with real-time data

OEE & Shop Floor:

- Overall Equipment Effectiveness calculation
- Availability, performance, quality metrics
- Shift handoff reports
- Digital work instructions
- Mobile shop floor interface with voice commands

Compliance Integration:

- ISO 13485 (Medical), AS9100D (Aerospace), IATF 16949 (Automotive)
- API Standards (Oil & Gas), SOC2, HIPAA, FDA 21 CFR Part 11
- Compliance-as-Code with auto-provisioning and gap analysis
- Append-only audit logs

Subscription Tiers (Built-in SaaS Model):

- Essentials → Professional → Enterprise tier definitions
- Per-tenant billing configuration
- Feature gating per tier
- Usage tracking and metering

Estimated Value: \$200K-\$400K/yr per enterprise Savings: Replaces JobBOSS/E2/ProShop (\$15K-\$50K/yr) + QuickBooks (\$5K-\$15K/yr) + separate CRM (\$3K-\$10K/yr) + scheduling software (\$5K-\$15K/yr) + inventory system (\$3K-\$10K/yr). Total software replacement: \$31K-\$100K/yr. Labor savings from automation: 2-3 FTE equivalent (\$80K-\$180K/yr).

COMPLETE PLATFORM INVENTORY (Beyond the 4 Flagships)

Summary Statistics

Metric	Count
MCP Dispatchers	67
Total Actions	2,474+
Calculation Engines	880+
Algorithms	52 (safety-classified, peer-reviewed)
Formulas	499 across 27 domains
Data Registries	22
Materials Database	6,346+ (127 parameters each)
Tool Catalog	95,608+ tools (28+ manufacturers)
Machine Profiles	888 (43 manufacturers, Level 4-5)
Controller Families	12
Hooks	157+ (1,107 trigger events, 44 safety-critical)
Skills/Commands	318+ skills, 163+ scripts
Agent Definitions	70 (Opus/Sonnet/Haiku routed)
API Endpoints	322 (32 route modules)
Compliance Frameworks	7
Toolpath Strategies	697+ cross-CAM
University Courses	8+ (MIT/Stanford)

Additional Feature Domains

Safety & Machine Protection (30+ Actions, SAFETY CRITICAL)

- Multi-layer collision detection, coolant validation, spindle protection
- Tool breakage prediction, workholding safety, 44 non-disableable blocking hooks
- **Value: \$100K-\$200K/yr** | Prevents \$50K-\$500K crash events

Quality & Metrology (13+ Actions, 43 Engines)

- Full SPC (Cp, Cpk, EWMA, CUSUM), CMM planning, tolerance stack-up, GD&T validation
- Gauge R&R, Monte Carlo uncertainty, probing programs
- **Value: \$80K-\$120K/yr** | Replaces Minitab/InfinityQS (\$10K-\$30K/yr)

Predictive Maintenance & Diagnostics (20+ Actions)

- Failure prediction, tool wear monitoring (Archard/Usui), chatter detection
- Thermal management, failure forensics, digital twin sync
- **Value: \$80K-\$150K/yr** | Reduces unplanned downtime 30-50%

Real-Time Machine Monitoring (40+ Actions)

- OPC-UA connectivity, live status, adaptive feed/spindle control
- Bayesian calibration, Grafana/Prometheus, mobile interface
- **Value: \$120K-\$200K/yr** | Replaces MachineMetrics (\$500-\$2K/machine/yr)

AI/ML & Neural Intelligence (35+ Engines)

- Federated learning, knowledge graph, manufacturing genome
- NLP-to-CAM parsing, self-learning hooks, RuVector intelligence
- **Value: \$150K-\$300K/yr** | 10-30% improvements humans miss

Video Learning & Vision AI (4 Engines)

- Video to Ffmpeg to Claude Vision to CadQuery to STEP/STL
- Watch a tutorial video, get a 3D model and structured knowledge out
- **Value: \$80K-\$120K/yr** | Captures retiring machinist knowledge

Data Redaction & Privacy (Federated Learning Engine)

- Auto-redaction of shop name, customer, pricing, serials for cross-shop sharing
- Privacy scoring, anonymization reports, safe-to-share validation
- **Value: \$40K-\$60K/yr** | Enables federated learning without IP exposure

CAD Engine (57 Actions, 34 Engines)

- Parametric geometry, feature recognition, DFM analysis, assembly modeling
- Fusion 360 Live Bridge, CadQuery code gen, Print-to-Geometry
- **Value: \$120K-\$180K/yr**

Knowledge Management (18+ Actions)

- Shop practice KB, machining playbook, document learning, tribal knowledge capture
- **Value: \$60K-\$100K/yr** | Reduces new hire ramp-up from 6 months to 6 weeks

Multi-Tenant Platform (15 Actions)

- Tenant isolation, Shared Learning Bus, API gateway, OAuth 2.1 PKCE, RBAC
- **Value: \$100K-\$200K/yr** | Enables SaaS deployment

Sustainability (10 Actions, 9 Engines)

- Energy optimization, carbon footprint, coolant lifecycle, waste minimization
- **Value: \$40K-\$60K/yr**

UNIVERSITY ACADEMIC KNOWLEDGE BASE

Peer-reviewed, scientifically validated algorithms — not empirical guesses.

Course	University	Algorithms Used In PRISM
6.003	MIT	FFT, digital filters, spectral analysis (chatter detection)
2.14	MIT	PID control, state estimation (adaptive feed/spindle)
2.830J	MIT	Kienzle force, Taylor tool life, Merchant shear, chip mechanics
3.22	MIT	Johnson-Cook constitutive, fatigue, creep, fracture
6.041	MIT	Bayesian wear models, uncertainty quantification
18.06	MIT	Matrix ops, eigenvalues, SVD (vibration analysis)
CS234	Stanford	Reinforcement learning (adaptive control, RL post-processor)
ME 353	Various	ODE solvers, thermal FEA, surface integrity

New courses integrated in 1-2 days. Algorithm registry with safety classification makes expansion systematic and safe.

PLATFORM GENERALIZATION POTENTIAL

60-70% of PRISM is domain-agnostic infrastructure. Each new vertical needs only 30-40% new engines.

Target Domain	Reusable Components	New Engines	Time
Civil Engineering	Compliance, FEA, materials, scheduling, cost	Structural, soil, concrete	3-4 mo
Electrical Eng.	Simulation, signal processing, thermal, BOM	Circuit, PCB, EMC	3-4 mo
Architecture	CAD, BOM, cost, compliance, 3D viz	Building codes, HVAC	2-3 mo
Medicine	ISO 13485 (built), FDA (built), stats	Clinical, device testing	3-5 mo
Oil & Gas	API (built), NDT, maintenance, safety	Reservoir, drilling	3-4 mo
Additive Mfg	Material science, thermal, quality, cost	Melt pool, scan strategy	2-3 mo

Each vertical = \$1M-\$3M additional market opportunity.

MACHINE DATABASE — 888 Machines, 43 Manufacturers

Manufacturer	Count	Specialty
Haas	139	Popular VMC/HMC/lathe, USA
Mazak	107	INTEGREX, VARIAXIS, multi-tasking
DMG MORI	94	Multi-tasking, universal milling
Okuma	60	OSP control, Thermo-Friendly
DN Solutions	56	Full-line turning & milling
Hurco	49	Conversational CNC
Makino	30	High-speed die/mold, aerospace
Brother	28	High-speed drill/tap
Hyundai WIA	26	Full-line Korean precision
Matsuura	22	5-axis, multi-pallet
23 more mfgs	277	Hermle, Chiron, Kern, Kitamura, Citizen, etc.

Full CSV list attached separately: **PRISM_Machine_Database_Full_List.csv** (888 machines with specs)

TOTAL VALUE ASSESSMENT

Feature Domain	Low Est.	High Est.
Product 1: Ultimate Speed & Feed Calculator	\$120K	\$200K
Product 2: Post Processor Generator	\$200K	\$350K
Product 3: Print-to-Program Pipeline	\$150K	\$250K
Product 4: Full ERP/Business/Accounting	\$200K	\$400K
Safety & Machine Protection	\$100K	\$200K
Quality & Metrology	\$80K	\$120K
Predictive Maintenance	\$80K	\$150K
Real-Time Monitoring	\$120K	\$200K
AI/ML Intelligence	\$150K	\$300K
Video Learning & Vision AI	\$80K	\$120K
CAD Engine	\$120K	\$180K
Knowledge Management	\$60K	\$100K
Data Redaction & Privacy	\$40K	\$60K
Multi-Tenant Platform	\$100K	\$200K
Sustainability	\$40K	\$60K
Compliance (7 frameworks)	\$60K	\$100K
Materials Intelligence	\$60K	\$100K
TOTAL ANNUAL VALUE	\$1,960K	\$3,390K
Development Replacement Cost	\$8M	\$15M

Competitive Comparison

Competitor	What They Offer	Our Advantage
Xometry (\$1.2B mkt cap)	Upload-to-quote only	We do quote + program + monitor + manage
MachineMetrics (\$80M+ raised)	Machine monitoring only	We add physics, adaptive control, digital twin
ProShop ERP (\$50K+/yr)	Shop management only	We integrate manufacturing intelligence into ERP
Harvey/Sandvik Calculators	Empirical S/F lookup	We use Kienzle/Taylor physics with 95K tools

Autodesk HSMWorks Post	Single-CAM post processor	We do cross-CAM with custom algorithms + auto S/F
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PRISM combines ALL capabilities into a single AI-native platform with physics-based intelligence none of them offer.

Identified Gap

Gap	Priority	Effort	Impact
Digital Blueprint Redaction	HIGH	1-2 days	Safe RFQ print sharing